Kernix 0.7.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

14 workloads

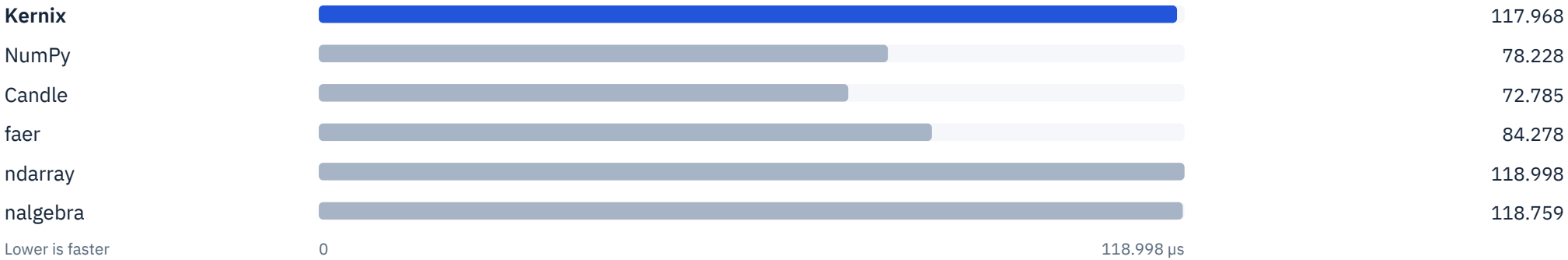
50 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

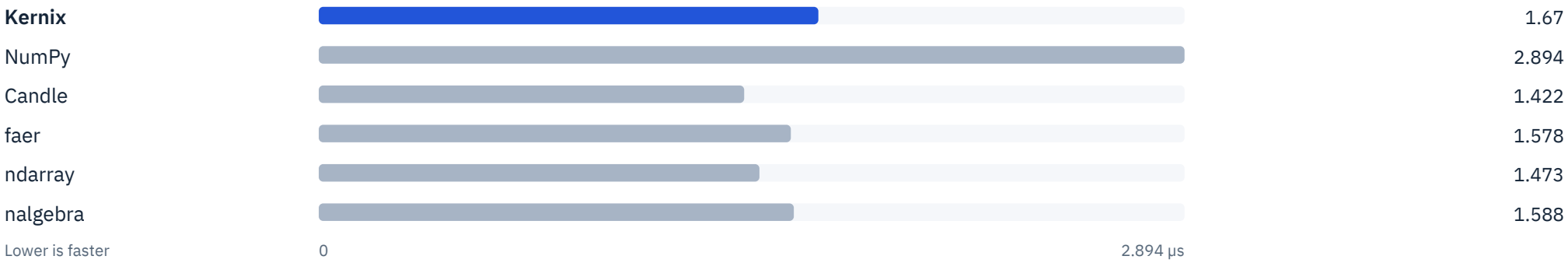
Kernix 0.7.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

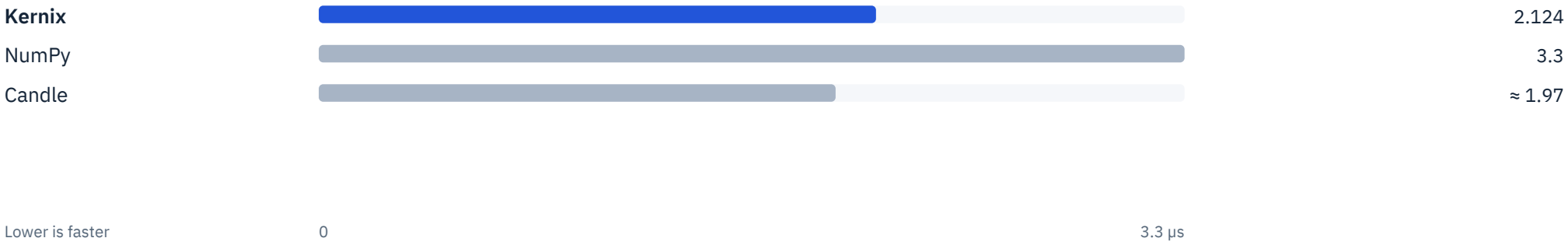
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



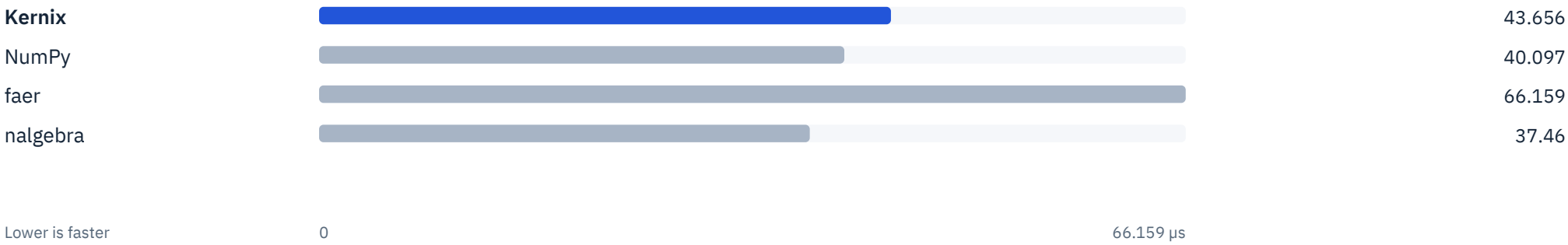
Kernix 0.7.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

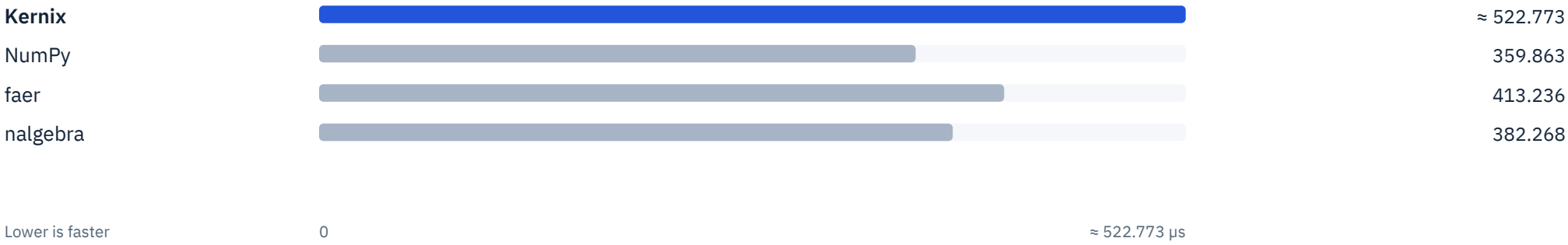
Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median

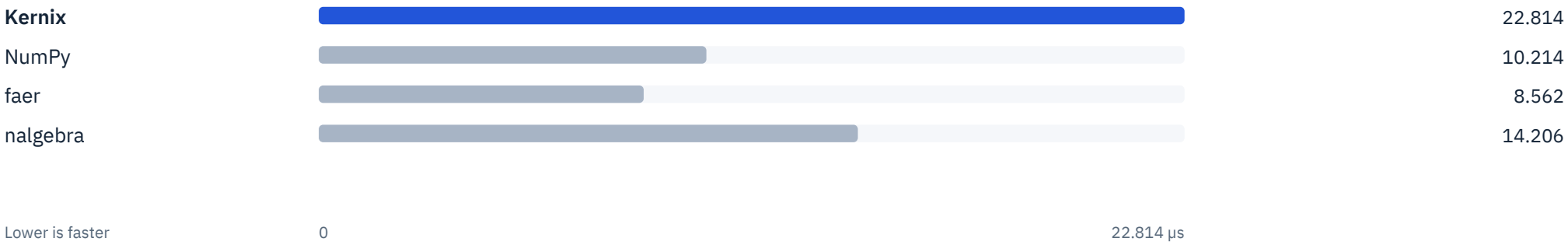


Kernix 0.7.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



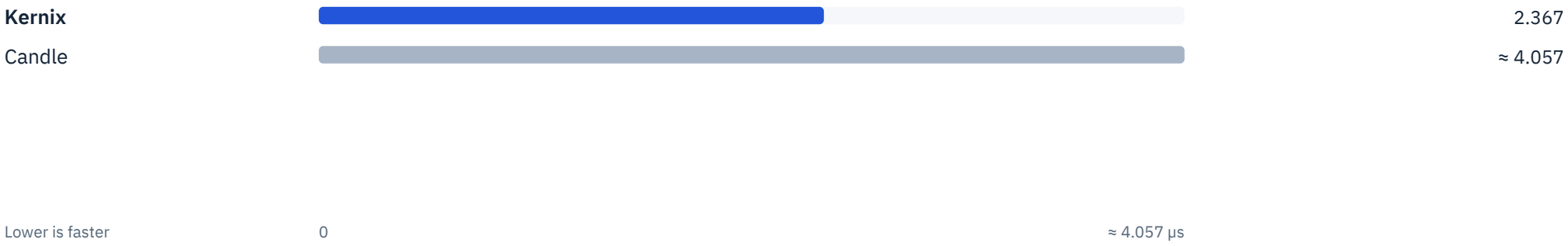
Kernix 0.7.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.7.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

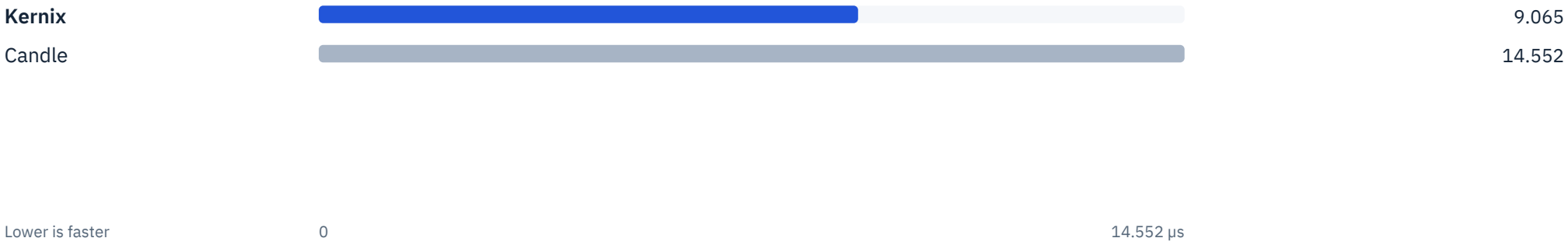
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.7.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

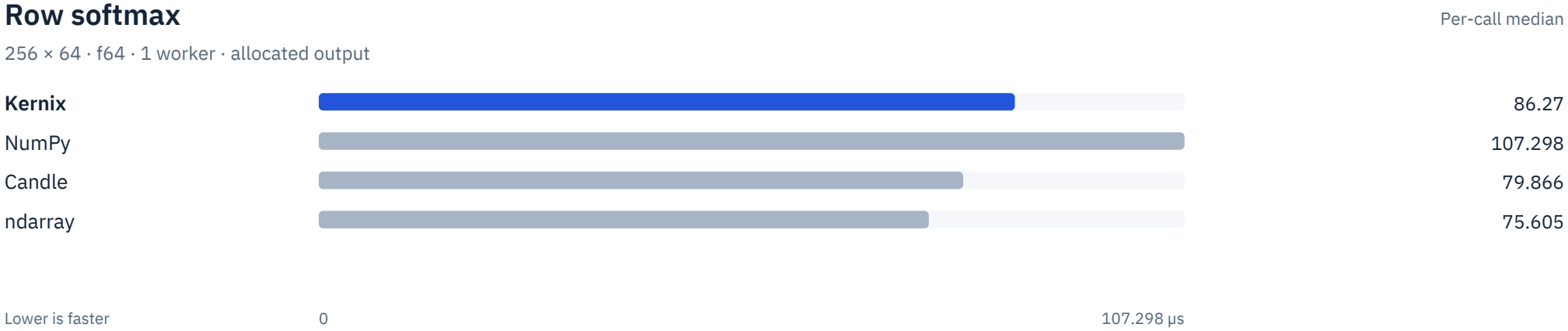


Kernix 0.7.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



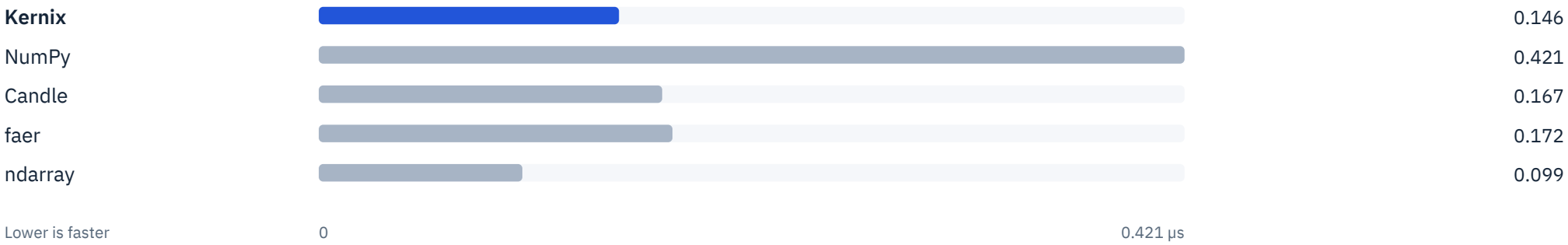
Kernix 0.7.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1670	1.67
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1473	1.473
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1588	1.588
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1578	1.578
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1422	1.422
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2894	2.894
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	117968	117.968
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	118998	118.998
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	118759	118.759
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	84278	84.278

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	72785	72.785
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	78228	78.228
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2124	2.124
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	1970.0000000000002	1.9700000000000002
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3300	3.3
Thin QR / Kernix 96 × 32 · f64	43656	43.656
Thin QR / nalgebra 96 × 32 · f64	37460	37.46
Thin QR / faer 96 × 32 · f64	66159	66.159
Thin QR / NumPy 96 × 32 · f64	40097	40.097
Thin QR / Kernix 256 × 64 · f64	522773.00000000006	522.77300000000006

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 256 × 64 · f64	382268	382.268
Thin QR / faer 256 × 64 · f64	413236	413.236
Thin QR / NumPy 256 × 64 · f64	359863	359.863
Cholesky factorization / Kernix 64 × 64 · f64	22814	22.814
Cholesky factorization / nalgebra 64 × 64 · f64	14206	14.206
Cholesky factorization / faer 64 × 64 · f64	8562	8.562
Cholesky factorization / NumPy 64 × 64 · f64	10214	10.214
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2367	2.367
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4057.0000000000005	4.0570000000000005
Model inference / Kernix batched normalized/sliced MLP · f64	4364	4.364

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	3801	3.801
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4864	4.864
Model recording + forward / Candle batched normalized/sliced MLP · f64	5214	5.214
Model backward / Kernix prepared graph → fresh gradients · f64	9065	9.065
Model backward / Candle prepared graph → fresh gradients · f64	14552	14.552
Parameter update / Kernix transactional SGD, fixed gradients · f64	481.00000000000006	0.48100000000000006
Row softmax / Kernix 256 × 64 · f64	86270	86.27
Row softmax / ndarray 256 × 64 · f64	75605	75.605
Row softmax / Candle 256 × 64 · f64	79866	79.866
Row softmax / NumPy 256 × 64 · f64	107298	107.298

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	146	0.146
Create zeros / ndarray 64 × 64 · f32	99	0.099
Create zeros / faer 64 × 64 · f32	172	0.172
Create zeros / Candle 64 × 64 · f32	167	0.167
Create zeros / NumPy 64 × 64 · f32	421	0.421
Create ones / Kernix 64 × 64 · f32	186	0.186
Create ones / ndarray 64 × 64 · f32	160	0.16
Create ones / faer 64 × 64 · f32	179	0.179
Create ones / Candle 64 × 64 · f32	232	0.232
Create ones / NumPy 64 × 64 · f32	1332	1.332